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Review Article

A comprehensive survey on weed and crop classification using machine learning and deep learning

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ABSTRACT

Machine learning and deep learning are subsets of Artificial Intelligence that have revolutionized object detection and classification in images or videos. This technology plays a crucial role in facilitating the transition from conventional to precision agriculture, particularly in the context of weed control. Precision agriculture, which previously relied on manual efforts, has now embraced the use of smart devices for more efficient weed detection. However, several challenges are associated with weed detection, including the visual similarity between weed and crop, occlusion and lighting effects, as well as the need for early-stage weed control. Therefore, this study aimed to provide a comprehensive review of the application of both traditional machine learning and deep learning, as well as the combination of the two methods, for weed detection across different crop fields. The results of this review show the advantages and disadvantages of using machine learning and deep learning. Generally, deep learning produced superior accuracy compared to machine learning under various conditions. Machine learning required the selection of the right combination of features to achieve high accuracy in classifying weed and crop, particularly under conditions consisting of lighting and early growth effects. Moreover, a precise segmentation stage would be required in cases of occlusion. Machine learning had the advantage of achieving real-time processing by producing smaller models than deep learning, thereby eliminating the need for additional GPUs. However, the development of GPU technology is currently rapid, so researchers are more often using deep learning for more accurate weed identification.
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1. Introduction

The global population is steadily growing each year, and according to world statistical data of United Nations Department of Economic and Social Affairs, Population Division, 2022, it is projected to increase above 8 billion in 2022. This increase in population presents several challenges, particularly in the aspect of food resource availability. Furthermore, the demand for daily food ingredients is projected to increase (Talaviya et al., 2020), as more individuals require sustenance, water, and other agricultural resources to fulfill nutritional requirements. Consequently, the agricultural sector is required to address the daily needs of the growing population, which is also a primary source of income for farmers. However, farmers encounter difficulties in producing agricultural products due to the diminishing agricultural areas caused by the rise in residential constructions. Farmers are also faced with the dual challenge of ensuring the quality of crop and increasing production. A significant challenge in achieving these goals is the presence of weed, which compete with surrounding plants (Berquer et al., 2023; Nathalie et al., 2020; Sabzi et al., 2020).

Weed grows rapidly in unwanted locations, such as gardens, fields, and agricultural areas. Weed absorb essential nutrients including sunlight, water, and soil space from agricultural plants (Kubiak et al., 2022; Raja et al., 2020a). Consequently, farmers need to adopt various methods to eradicate weed around crop. For example, farmers can resort to using pesticides (Monteiro and Santos, 2022) or manually cutting the weed with tools (Scavo and Mauromicale, 2020). Both approaches have drawbacks, as pesticides may adversely affect crop when not accurately targeted, leading to residues and wastage (Kalyabina et al., 2021). Meanwhile, manual cutting consumes farmers' time and energy, resulting in inefficiency. The recognition of these challenges causes studies to actively explore alternative methods or technologies for weed control (Liu and Bruch, 2020). Integrating

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technology into weed removal processes can aid farmers in reducing manual workloads, enhancing efficiency, and achieving better crop yields.

Aside from enhancing efficiency, the utilization of automation for weed removal brings about additional advantages, including cost-effectiveness and environmental friendliness. An example is the ability of farmers to conserve labor while effectively clearing weed (Rajmis et al., 2022). However, it is crucial to be aware that inaccurate pesticide application can lead to environmental pollution (Zaller et al., 2022). The environmental pollution of pesticides arises from their chemical properties, which contaminate water, soil, air, and the surrounding ecosystem. Consequently, the development of an automated weed-spraying system necessitates a meticulous classification process for both weed and crop (Liu and Bruch, 2020). On the other hand, accurate weed and crop categorization present unique challenges due to the uniform green color of leaves, which is further complicated by variations in color caused by lighting conditions (Kazerouni et al., 2019). The growth of weed often matches with that of crop, resulting in occlusion between leaves (L. Zhang et al., 2022b). Another factor is the similarity in size between the leaves of small crop and weed (Chen et al., 2020). Fig. 1 shows the frequent challenges faced when classifying weed and crop.

Numerous previous studies have explored the classification of weed and crop. An example is the works of Hasan et al., 2021 and Al-Badri et al., 2022b, which reviewed the application of deep learning and traditional machine learning for classifying weed. Both machine learning and deep learning were instrumental in efficiently and accurately identifying weed and crop. These technologies minimize human effort and improve crop management, particularly in the context of identifying weed and crop (Vasileiou et al., 2023). Typically, the main difference between machine learning and deep learning lies in the model complexity (Shah and Bhavsar, 2022; Siemers et al., 2022). Machine learning focuses on handcrafting algorithms and statistical models that enable computers to learn from data (Moshawrab et al., 2023), thereby facilitating predictions or decisions based on discerned patterns (Pandey et al., 2023). Meanwhile, deep learning mirrors the structure of the human brain (Bai et al., 2021), with interconnected neurons arranged in layers. This approach allows the model to learn automatically from the data by extracting features hierarchically (Banan et al., 2020) and producing an increasingly complex representation as the depth of the neural network layer increases (Samanta et al., 2023). Machine learning excels in simpler tasks with a limited number of features and uses less complex mathematical calculations, making it suitable for Internet of Things (IoT) devices (Phasinam and Kassanuk, 2022; Phasinam et al., 2022). On the other hand, deep learning is more effective in handling complex, data-intensive tasks, particularly those incorporating visual pattern recognition.

Previous survey papers have explored individual aspects of machine learning (Venkataraju et al., 2023; A. Wang et al., 2019; Z. Wu et al., 2021a) or deep learning (Hasan et al., 2021; K. Hu et al., 2024; Murad et al., 2023), but none have offered a comprehensive discussion of both in the context of this specific case study. Therefore, this review aimed to bridge the existing knowledge gap by conducting a comprehensive analysis of the various phases involved in machine learning and deep learning. It will include an in-depth exploration of the pre-processing, segmentation, feature extraction, and classification stages. The initial stage consists of pre-processing, where the dataset is prepared for subsequent processing. Following this process, segmentation is performed by separating leaf objects from the surrounding background, including soil and other elements. Subsequently, feature extraction is carried out to identify common features within the leaves. The final stage is classification, where weed and crop are categorized. At the end of this review, three challenges related to weed and crop detection are addressed, namely 1) the impact of lighting effects, 2) occlusion conditions, and 3) classification during the early growth phase. Additionally, an analysis is conducted on accuracy values and processing times, particularly in real-time applications. Implementations have been tested on Raspberries, computers (laptops), and robots. The primary goal of this review is to assist readers in navigating the optimal methods, particularly when faced with the complex challenges caused by classification and real-time processing under three specific conditions.

Based on the arrangement of this study, Section 2 describes the necessity of reviewing the classification of weed and crop. Furthermore, Section 3 explores weed detection and its categorization methods. Section 4 discusses the classification of weed and crop using machine learning. Section 5 explains deep learning applications for classifying weed and crop. Section 6 addresses the utilization of hybrid convolutional networks and machine learning methods in classification case studies. Section 7 also benchmarks the dataset to measure accuracy and speed. Section 8 contains future work and discussions, which explains problems commonly encountered during investigation. Finally, Section 9 concludes by providing insights into methods suitable for the classification process of weed and crop.

2. Related surveys

Several studies extensively used machine learning and deep learning for the detection and identification of weeds and crops, primarily focusing on weed control. Additionally, several review articles have delved into the utilization of these technologies, focusing particularly on either maximizing accuracy or enhancing the efficiency of processing speeds. In 2021, Hasan et al., 2021 conducted a comprehensive survey of deep learning methods for weed detection from image data, analyzing 70 papers across data acquisition, pre-processing, detection and classification, and evaluation. The result showed that fine-tuning deep learning models had very high classification accuracy, but a limitation was recorded in the form of computational speed. Another survey by Murad et al., 2023 explored the use of deep learning for weed detection, primarily using RGB-format datasets. To achieve optimal accuracy, Convolutional Neural Network (CNN) architectures were modified, but the survey showed extended training times, necessitating GPU devices. Additionally, K. Hu et al., 2024 contributed to the discourse by

(a) Light illumination (Kazerouni et al., 2019) (b) Occlusion on leaves (L. Zhang et al., 2022) (c) Early-stage growth (Chen et al., 2020)

Fig. 1. Challenges in a weed detection system.

reviewing the use of deep learning, focusing on datasets, evaluation methods, and weed recognition techniques. The survey identified occlusion as a challenge in weed detection. Despite the high accuracy of deep learning, the architecture introduced complexities during training and deployment.

Machine learning methods for classifying weed and crop were also carried out by Venkataraju et al., 2023, which reviewed 35 papers on weed detection in corn plants. The frequently used method in this context was Support Vector Machine (SVM) and Neural Network. Similarly, Z. Wu et al., 2021a explored machine learning methods for computer vision-based weed detection, comparing features such as texture, shape, color, and spectral characteristics. The combination of these features was instrumental in achieving accurate results. A. Wang et al., 2019 also conducted a comprehensive analysis of machine vision techniques enhanced by machine learning, including steps such as pre-processing, segmentation, feature extraction, and classification. Their research delved into four main categories of features, namely biological morphology, spectral characteristics, visual textures, and spatial context. This integration of machine learning with image processing demonstrated promising capabilities for accurate and real-time identification of weeds and crops.

Despite several previous survey papers, a comprehensive review addressing the strengths and weaknesses of both machine learning and deep learning in the context of weed detection is lacking. Furthermore, an in-depth discussion on pre-processing stages, segmentation, feature extraction, and classification for weed and crop is presented. The exploration extends to common challenges in weed and agricultural plant detection, comprising the impact of light effects, occlusion conditions on leaves, the early-growth plant phase, and real-time processing for future field applications.

3. Weed control and categorization method

Weed consists of various forms, generally categorized as narrow-leaf and broad-leaf as shown in Fig. 2. The economic impact of weed is significant, leading to reduced crop yields (Kubiak et al., 2022), increased production costs (Sharma et al., 2021), and reduced crop quality (Monteiro and Santos, 2022). Consequently, effective weed control is an essential component in agriculture to sustain crop productivity and profitability. Currently, two predominant trends correspond to distinct methodologies for weed control in agricultural systems. While synthetic herbicides are prevalent in the conventional approach (Frimpong et al., 2018), the modern approach leans towards using mechanical, precision agriculture, and sensor-based strategies (Machleb et al., 2020).

Conventional weed control comprises manual labor, cultural practices, and herbicide application. Specifically, manual labor consists of the physical removal of weed by hand (Woyessa et al., 2022), a labor-intensive yet effective method suitable for small-scale farming. Cultural practices such as crop rotation, tillage, and mulching disrupt weed growth and prevent seed germination. Additionally, herbicides play a key role in conventional weed control with chemical substances applied to the soil or plant foliage for weed control (Strehlow et al., 2020). Herbicides can be selective, targeting only specific weed types, or non-selective, which affects all plants. While herbicides have revolutionized weed management, enabling large-scale control and increasing crop yields, overreliance can result in herbicide-resistant weed (Gaines et al., 2020), environmental pollution (S. Zhang et al., 2021), and health hazards for farmers and farmworkers (Shammi et al., 2020).

Modern weed control combines traditional methods with innovative technologies such as precision agriculture (Kent Shannon et al., 2018), remote sensing (Farooq et al., 2018b), and robotics (Raja et al., 2020b). Precision agriculture incorporates the use of sensors, GPS, and computer vision for site-specific weed management, reducing herbicide usage and enhancing crop yields. Meanwhile, remote sensing technologies, including satellite imagery (Rasmussen et al., 2021) and Unmanned Aerial Vehicles (UAVs) (Mohidem et al., 2021), detect and map weed infestations, which enables targeted control measures. Robotics, particularly autonomous robots equipped with computer vision and machine learning algorithms, offer real-time weed detection and removal, reducing reliance on manual labor and herbicides. It's essential to recognize the transformative impact that these technologies may have on weed control. They offer a pathway towards a more efficient, sustainable, and eco-friendly method of controlling weeds.

Modern weed control is closely associated with Artificial Intelligence (W.-H. Su, 2020; Subeesh and Mehta, 2021). According to Fig. 3, weed detection is categorized into traditional machine learning, deep

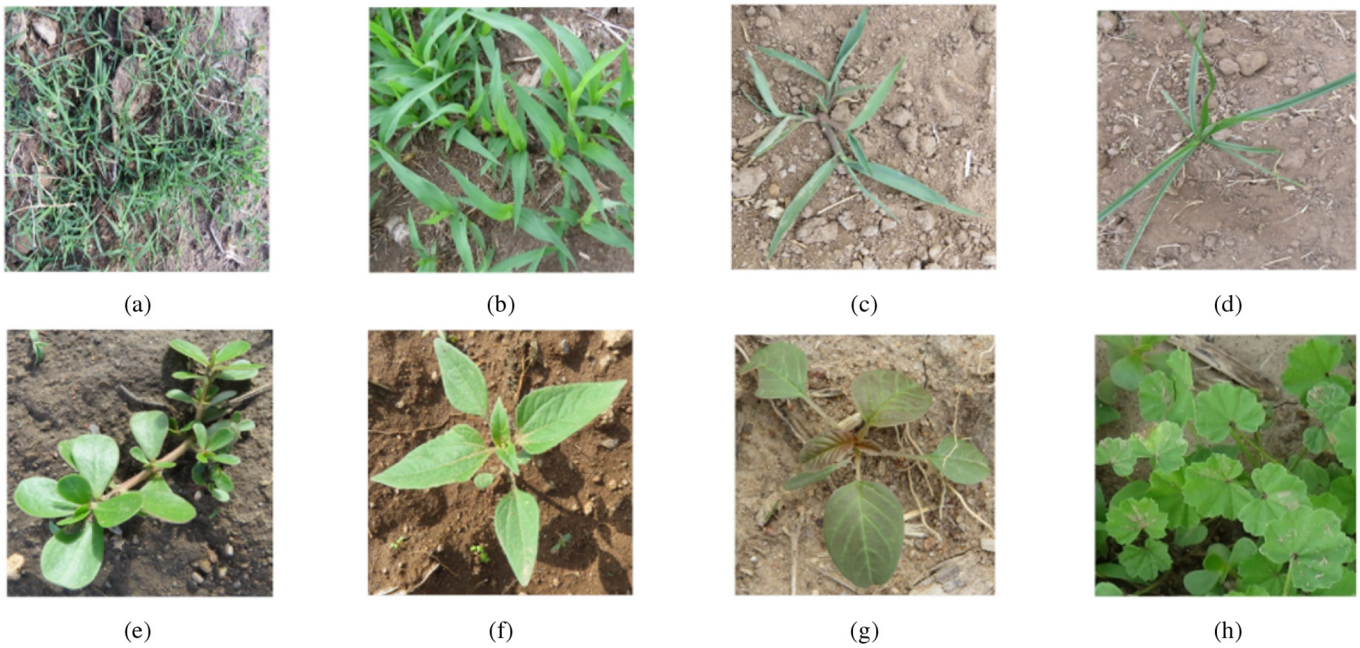

Fig. 2. Sample of narrow-leaf weeds: (a) *Cynodon dactylon*. (b) *Eleusine indica*. (c) *Digitaria sanguinalis*. (d) *Cyperus esculentus* and broadleaf weeds: (e) *Portulaca oleracea*. (f) Tithonia tubaeformis (Jacq.) Cass. (g) Amarantus spinosus. (h) *Malva parviflora*. (Garibaldi-Márquez et al., 2022).

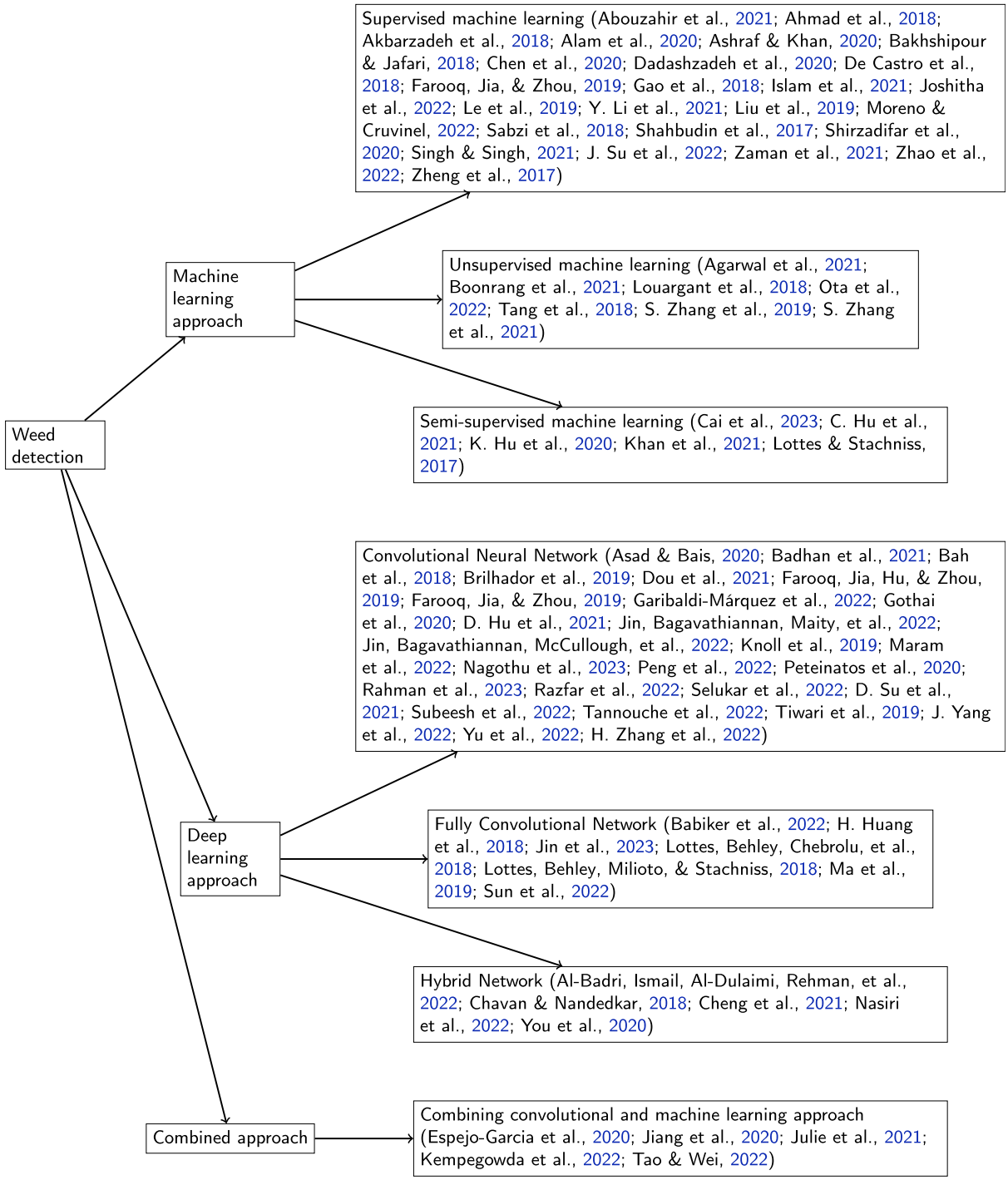

Figure 3: Weed detection categories

et al., 2021; Zhao et al., 2022). It is crucial to acknowledge that optimal feature combinations can further enhance accuracy in weed and crop detection (L. Zhang et al., 2022). Consequently, this review provides examples of features as input to traditional machine learning methods, showing the importance of accurately classifying weed and crop species.

Deep learning approaches are further categorized into CNN, Fully Convolutional Networks (FCN), and Hybrid Networks. CNN model typically comprises two fundamental

Fig. 3. Weed detection categories.

learning, and a combined approach. Machine learning techniques are categorized into three main groups, namely supervised learning, unsupervised learning, and semi-supervised learning. Supervised learning is labeled training and validation datasets (Chegini et al., 2019), where both the image and its accompanying labels are part of the input dataset for the machine learning model. However, unsupervised learning operates with an unlabeled training set, and the input dataset for the unsupervised model lacks annotations (J. Wang and Biljecki, 2022). This method enables the model to map the input to a specific output. Meanwhile, Semi-supervised learning occupies a middle ground as it is positioned between supervised and unsupervised learning (Chang et al., 2020). The utilization of traditional machine learning methods in weed and crop classification offers several advantages. Typically, traditional machine learning requires only a limited number of datasets for training, resulting in a relatively small model size. This compact model facilitates swift recognition of leaf objects and allows for real-time operation. Additionally, the features used in traditional machine learning are not mathematically complex. These characteristics make the method to be well-suited for real-time weed and crop detection (Tufail et al., 2021; Zhao et al., 2022). It is crucial to acknowledge that optimal feature combinations can further enhance accuracy in weed and crop detection (L. Zhang et al., 2022b). Consequently, this review provides examples of features as input to traditional machine learning methods, showing the importance of accurately classifying weed and crop species.

Deep learning approaches are further categorized into CNN, Fully Convolutional Networks (FCN), and Hybrid Networks. CNN model typically comprises two fundamental components, namely feature extraction and classification (Indolia et al., 2018). Various studies have implemented numerous CNN models, including VGGNet (Simonyan and Zisserman, 2014), DenseNet (G. Huang et al., 2017), MobileNet (Howard et al., 2017), ResNet (Shafiq and Gu, 2022), etc. These models incorporated diverse feature extraction and classification layers. Compared to CNN, FCN modifies the architecture by substituting the fully connected layers with convolutional ones and employs a transposed convolution layer to generate an output image that matches the input size in dimensions (J. Wu et al., 2021b). Additionally, hybrid architectures have been developed by combining the characteristics of two or more deep learning models (X. Zhang et al., 2020). For example, the hybrid approach combines convolutional feature extraction with machine learning as a classifier. It is essential to be aware that both machine learning and deep learning comprise data acquisition processes.

This review classifies images into three types, namely hyperspectral images (Farooq et al., 2019a; Farooq et al., 2019b; Farooq et al., 2018a; Gao et al., 2018; Y. Li et al., 2021), multispectral images (Farooq et al., 2019a; Liu et al., 2019; Louargant et al., 2018; J. Su et al., 2022), and digital images (Chen et al., 2020; Rahman et al., 2023; S. Zhang et al., 2021). Hyperspectral images are captured using sensors capable of collecting information across adjacent spectral bands. Hyperspectral imaging sensors often use a greater number of narrower bands to capture more specific spectrum information. The entire spectrum in each pixel of hyperspectral images has been applied in various agricultural contexts. However, a multispectral camera is suitable for examining larger areas of interest, as it is more portable and cost-effective with higher spatial resolution. Additionally, the multispectral camera used is equipped with more spectral bands and is less susceptible to environmental changes due to the availability of a reflectance calibration panel. Digital sensors or cameras produce digital images, which can be either color or black-and-white, with spatial resolution contingent on the sensor or camera quality. In general, the choice of image type also influences the input dataset used.

4. Weed detection using machine learning

Weed detection using image or video data leads to classification, which comprises distinguishing between species or differentiating weed from crop. In the context of machine learning, a crucial step is plant segmentation (Kitzler et al., 2022; Sakeef et al., 2023). This stage is very important for separating the background from the plants to be classified (K. Yang et al., 2020). Additionally, the features used for the classification process should be specifically related to the plant area. Land areas and other non-plant objects are typically removed or masked in black. Machine learning methods often use handcrafted features (Ghazal et al., 2021), manually designed and extracted from the object image based on knowledge to extract crucial information. In computer vision, these handcrafted features describe visual characteristics including color, texture, and shape (C. Yang, 2021)., capturing specific visual properties relevant to the case study, such as weed detection. The strengths and weaknesses of the traditional machine learning method are shown in Table 1. Machine learning models commonly follow the process shown in Fig. 4. Generally, machine learning consists of five stages, namely image acquisition, pre-processing, image segmentation, feature extraction, and classification. However, an image segmentation stage typically precedes the image feature extraction stage for enhanced precision (Ahmad et al., 2018). Typically, the segmentation stage is needed for preparing visual data for further analysis. By reducing data complexity, segregating objects from the background, and facilitating more effective feature extraction, segmentation contributes to constructing a focused and relevant data representation of the object under investigation.

4.1. Pre-processing

Pre-processing stage incorporates preparing images before proceeding to further processing. The objective of this process is to improve image quality, thereby enabling smoother analysis and processing. In this study, pre-processing primarily consists of image enhancement procedures including color conversion, which includes transitioning RGB colors to grayscale (Joshitha et al., 2022), using HSV colors (Agarwal et al., 2021; Moreno and Cruvinel, 2022), and adopting specific color channels, such as green (Ashraf and Khan, 2020). Fig. 5b shows an example of image pre-processing, where color conversion is used to streamline computational processes or prepare the image for subsequent stages. Another pre-processing step focuses on segmentation, particularly focusing on green-colored plants. Additionally, image resizing is implemented to simplify computations and standardize dataset sizes (Badhan et al., 2021; Shahbudin et al., 2017). Image smoothing is another pre-processing step carried out by using a Gaussian filter in order to reduce noise in the image (Rasmussen et al., 2021). Subsequently, the results of data pre-processing are used in the next processes, such as image segmentation or feature extraction.

4.2. Image segmentation

Segmentation is the primary step that needs to be taken to classify weed and crop, and it requires the isolation of regions of interest and the exclusion of non-relevant areas from the image. Typically, color is used to differentiate leaf objects (S. Zhang et al., 2021), and in most instances, weed or crop appear green, while soil is yellowish-brown or reddish, and stubble is yellowish. Using the distinctive color qualities of these elements is a common and straightforward method for segregating plants in the field. For example, Moreno and Cruvinel, 2022 applied the HSV color model, specifically using the green hue for segmentation, as shown in Fig. 5.

Segmentation comprises distinguishing between the foreground (crop) and background (weed) by dividing the image based on shared characteristics namely soil, rocks, and residue. Furthermore, segmentation process can also incorporate clustering, such as the K-means clustering approach (Ota et al., 2022; Tang et al., 2018), which categorizes pixels into K number of classes based on attributes. This approach is particularly useful for leaf weed segmentation, resulting in classified leaf area images.

Table 1
Strengths and weaknesses of research with the traditional machine learning.

Researcher	Method	Strength	Weakness
Supervised learning			
Gao et al. (2018)	Normalized difference vegetation index (NDVI), ratio vegetation index (RVI), Band reflectance, PCA, Random Forest	The use of the reduction feature can produce high precision and recall values.	Does not carry out the segmentation process so that there is still soil area.
Bakhshipour and Jafari (2018)	Green Minus Red (GMR), Shape feature, ANN or SVM	Shape features are informative for use in weed/crop discrimination at early stages of growth.	This method is only suitable for use in the four-leaf stage.
J. Su et al. (2022)	Five bands (Blue, Green, Red, RedEdge, NIR), Mutual Information and Sequential Forward Selection, Random Forest with Bayesian Optimization	Spatial information helps improve classification and reduce noise.	The segmentation stage is not carried out, so the UAV flight is limited to a certain height.
Islam et al. (2021)	Normalized red, green, blue, RF, SVM, KNN	Using an inexpensive RGB color camera can produce good accuracy.	A manual labeling process with coloring in the Region of Interest (ROI).
Y. Li et al. (2021)	HSI spectrograph, thresholding, superpixels, SVM, partial least squares-discriminant analysis, Multi-layer Perceptron	Hyperspectral imaging is capable of identifying the unique spectral patterns emitted by a wide variety of weed species.	Weed in a pot, not in a large field.
Zhao et al. (2022)	Image skeleton, ring structure, Support Vector Machine	The method can run in real-time at 38 ms per image.	Prediction error due to crop overlap.
Joshitha et al. (2022)	RGB to Gray, HOG transform, a statistical binary classifier	The accuracy results are very good, combining 80% training and 20% testing.	A few datasets are used for training, so the data will be less accurate if there is a variation.
Alam et al. (2020)	shape, textures, and color, Random Forest	The system can work in varying environmental conditions.	Real-time results of 57.4 ms per image using PC Core i5.
De Castro et al. (2018)	Digital Surface Models (DSMs), orthomosaics, Random Forest	Mapping weed growth over large areas of land.	The mosaicking image acquisition process and running algorithm took 40 h.
Chen et al. (2020)	Gray level-gradient co-occurrence matrix, rotation invariant local binary pattern (LBP), SVM	The method can be used for early detection, starting from seedlings.	The prediction speed is too long, 10,000 s
Shahbudin et al. (2017)	Gabor Wavelet, FFT, One-class SVM	The resulting accuracy is very good for distinguishing narrow and broad weed.	The data used is small, so the data will be less accurate if there is a variation.
Dadashzadeh et al. (2020)	Bee algorithm, Artificial Neural Network	The method can classify narrow and broad weed in cloudy conditions.	Total execution per frame 0.633 s.
Ahmad et al. (2018)	Edge Orientation Histogram, Shape Matrix Histogram, AdaBoost Naïve Bayes	The system can run in real-time (28.4 fps) with Core i5 and 320 × 240 resolution.	Classification only on two types of weed.
Zaman et al. (2021)	Statistical (mean and variance), Gabor filter, SVM, Multi-Layer Perceptron NN	The method is suitable for classification based on texture.	The computational process is a little complex, so it cannot be used for real-time processing.
Singh and Singh (2021)	HSI, Gray, GLCM, Multi-Layer Perceptron NN	The use of color and texture helps the classification process	The data used is small, so if there is a variation, the data will be less accurate.
Ashraf and Khan (2020)	Green channel, Mean, variance, kurtosis, skew, complex moment, Random Forest	The proposed method can distinguish weed density.	The computational time of the feature extraction stage is very high.
Farooq et al. (2019b)	NIR spectrum, LBP-HOG, SVM	Use of shape and texture features in RGB images.	The resulting accuracy is not good for classifying two types of weed.
Moreno and Cruvinel (2022)	HSV with green Hue, shape, area, perimeter, width, height, proportion, rectangularity, circularity, Haralick moment, SVM	The use of the shape feature can distinguish weed and maize.	Overlapping leaves make the prediction results not optimal.
Zheng et al. (2017)	Color indices, PCA, support vector data description	A color-based method is less affected by camera focus, leaf orientation, wind, and canopy overlap.	Using the method in the leaf segmentation stage is a computational expense.
Akbarzadeh et al. (2018)	Gaussian SVM with raw data inputs	Able to classify broad-leaved (silver beet) and narrow-leaved (corn) plants accurately	Classification is only done on two types of plants.
Le et al. (2019)	Local Binary Pattern, Support Vector Machine	The method can run in real-time at 45.41 ms per image with a high computer hardware specification.	It is not easy to distinguish canola and radish leaves because of the similarity in shapes.
Abouzahir et al. (2021)	Bag-of-visual-words-augmented Histogram of Oriented Gradients, Back-propagation neural network	The method can detect dense and overlapped plants.	The method is not suitable for various illumination conditions.
Sabzi et al. (2018)	metaheuristic cultural algorithm, Hybrid Artificial Neural Networks – Harmony Search algorithm	The method is very accurate in outdoor light conditions.	The weakness of this method is the plant density due to overlapping.
Liu et al. (2019)	spectral bands and wavelet coefficients, Support Vector Machine	Adding the number of spectral bands can improve accuracy in the actual field environment of weed and crop.	The addition of spectral bands also adds to the complexity of the model.
Shirzadifar et al. (2020)	Spectral weed indices, Random Forest	This approach is suitable for determining sensitivity and resistance in greenhouse weed.	The dataset is limited to greenhouses with special treatment.
Unsupervised learning			
S. Zhang et al. (2021)	Modified Grabcut, Adaptive fuzzy dynamic K-means, Sparse representation classification	The segmentation stage avoids manual interaction, avoids dead zones, and redundancy centers on clustering.	The method can only be implemented on plant growth of 2–5 leaves.
S. Zhang et al. (2019)	Grabcut and K-Means Clustering, local weighted maximum discriminant	This method can reduce the dimensionality of the segmented weed image.	This method has a higher computational complexity.

	projection, Support Vector Machine		
Louargant et al. (2018)	Fourier transforms, NDVI threshold, Hough transform (spatial and spectral), SVM.	Datasets do not require labeling.	This method is not suitable for narrow weed.
Ota et al. (2022)	Blob feature, K-Means, Binary classification	This method overcomes the mislabelling of weed between crop and overlapping plants.	The precision of crop detection is still small due to the large size of weed, which results in shape similarity.
Agarwal et al. (2021)	Color spaces (RGB, CIELab, HSV), NDVI, VARI, K-Means	Color space can be used for clustering weed.	The time required for the color extraction process is 2 s, and 15 s for clustering.
Boonrang et al. (2021)	Mean shift, SVM	Clustering can remove noise in the image	The resulting accuracy is still small; the best is 82.6%.
Tang et al. (2018)	K-Means, SVM multiclassification, and Soft-max regression	This method overcomes the unstable identification in a complicated condition	Increasing the number of clusters increases the computational time.
Semi-supervised learning			
Khan et al. (2021)	Semi-supervised Generative Adversarial Network (SGAN)	Methods can label data that has not been labeled.	The method cannot run in real-time.
Lottes and Stachniss (2017)	RGB and NIR image, gradient, color space, texture, Random Forest	Fairly fast labeling time.	Processing time per image is quite long, which is 124 ms.
C. Hu et al. (2021b)	color match, training-time augmentation, and CNN with iterative semi-supervised learning	The method avoids being manually labeled and improves model performance.	The method cannot run in real time yet.
Cai et al. (2023)	Efficient Channel Attention Network on Spatial Pyramid Pooling (SPP) layer	This method improves the utilization of limited data and enhances model performance in complex environments.	The resulting accuracy could be better.
K. Hu et al. (2020)	Graph Weeds Net	This method makes it possible to find the image's key patches, which are accurate to target plants even when there are background and other plants.	The architecture still needs to be more efficient, so it is computationally complex.

4.3. Feature extraction

Features used in the classification process can be categorized into visual or visually observable features. Specifically, visual features, such as shape (Ahmad et al., 2018; Alam et al., 2020; Bakhshipour and Jafari, 2018; Farooq et al., 2019b; Moreno and Cruvinel, 2022), texture (Ahmad et al., 2018; Alam et al., 2020; Farooq et al., 2019b; Moreno and Cruvinel, 2022), and color (Alam et al., 2020; Islam et al., 2021; Moreno and Cruvinel, 2022; Zheng et al., 2017) have been considered by some studies. It is crucial to be aware that visual features play an important role in distinguishing between weed and crop. For example, in the case of corn and weed, despite both being green, their leaf shapes differ (Moreno and Cruvinel, 2022), particularly when at the early growth stage (Bakhshipour and Jafari, 2018; Islam et al., 2021). The advantage of visual features is the real-time applicability due to the algorithm's fast computation process (Ahmad et al., 2018; Alam et al., 2020). However, a significant drawback arises when there is occlusion on leaves covered by others (Moreno and Cruvinel, 2022) or when limited to a certain number of leaves' growth (Ahmad et al., 2018; Bakhshipour and Jafari, 2018). While the use of color indices is less affected by factors such as camera focus, leaf orientation, wind, and canopy overlap, it is computationally expensive when performing leaf segmentation (Zheng et al., 2017).

Apart from visual features, statistical features offer advantages that are independent of leaf color (Ashraf and Khan, 2020; Zaman et al., 2021). However, due to the combination of various statistical features, the developed method has a high computational time. Texture features can use methods such as Histogram of Oriented Gradients (HOG) (Abouzahir et al., 2021; Farooq et al., 2019b; Joshitha et al., 2022), Local Binary Pattern (LBP) (Chen et al., 2020; Farooq et al., 2019b; Le et al., 2019), Gray-Level Co-occurrence Matrix (GLCM) (Chen et al., 2020; Singh and Singh, 2021), and Gabor (Shahbudin et al., 2017) to automatically generate images through feature descriptors. It has been observed that the use of descriptors is more accurate than visual features (Joshitha et al., 2022; Shahbudin et al., 2017). For example, Chen et al., 2020 used a rotation invariant local binary pattern (LBP), as shown in Fig. 6, and a gray level-gradient co-occurrence matrix modified from Gray Level Co-Occurrence Matrix. Although this approach provides accurate results, its high complexity prevents real-time operation. On the other hand, Le et al., 2019 combined LBP operators to achieve real-time computer hardware speeds, but the use of a sophisticated computer impacts the resulting speed.

Visual features generally offer easily understandable information for human interpretation. For example, leaf shape, color, and texture information aid in identifying and distinguishing weed from other crop. Meanwhile, statistical features offer measurable information that can be mathematically calculated, such as analyzing the size and distribution of pixels in weed leaf images for precise quantitative results. Texture features are valuable for describing unique characteristics in order to produce descriptors capturing fine details from leaf images. However, in a real-time context, the selection of the right texture features is important, considering factors such as performance and execution speed.

4.4. Classification

Using supervised learning for the classification of weed and crop requires labeled datasets. Numerous investigations often utilize methods such as Artificial Neural Networks, Random Forests, Support Vector Machines, and Naïve Bayes for their research objectives. Typically, Neural network is a powerful tool in computer vision and general pattern recognition, with various parameters to configure, including the number of neurons, layers, functions, and input characteristics. Sabzi et al., 2018 used Neural Network to accurately classify a potato and three weeds, achieving results in less than 0.8 s on a standard computer. On the other hand, the Random Forest algorithm operates as a supervised

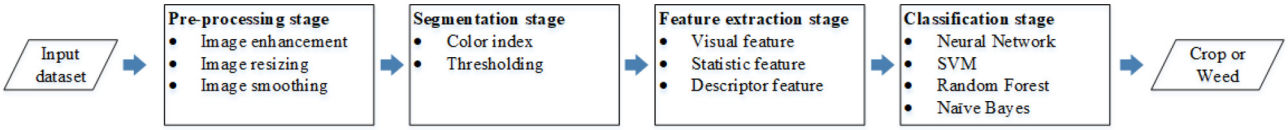

Fig. 4. A workflow of weed detection using machine learning.

Fig. 5. Segmentation process, a) original image, b) channel H of HSV color space converted image, c) thresholding on green value, d) remove small objects, e) final result (Moreno and Cruvinel, 2022).

machine learning technique suitable for addressing both classification and regression challenges. The method can construct multiple decision trees using randomly sampled data, thereby contributing to the model's predictive capabilities (Alam et al., 2020). Consequently, Random Forest consistently shows good classification accuracy (Gao et al., 2018; Islam et al., 2021; J. Su et al., 2022). For example, when applied with specialized treatment for plants, a 100% accuracy can be achieved (Shirzadifar et al., 2020).

Support Vector Machine (SVM) represents a form of supervised learning that formulates functions to map input to output based on labeled training data. This approach is aimed at predicting outcomes by learning from examples provided during training. Bakhshipour and Jafari, 2018 used SVM with shape features for weed and sugar beet detection, leveraging the informative nature of plant shape characteristics during early growth. However, advanced threshold-based classification features are essential due to variations in interference features between the primary crop and weed. The results of weed and sugar beet classification are shown in Fig. 7. Weeds are red, sugar beets are green, and yellow pixels are misidentified non-desirable objects. SVM is also effective across various datasets, including RGB data (Farooq et al., 2019b; Islam et al., 2021), plant seedling datasets (Chen et al., 2020), and differentiation between narrow and broad plants (Akbarzadeh et al., 2018; Shahbudin et al., 2017). The method also demonstrates real-time capability (Zhao et al., 2022). AdaBoost with Naïve Bayes is another supervised method capable of real-time operation (Ahmad et al., 2018). However, the models generated from supervised machine learning are often too small for real-time execution.

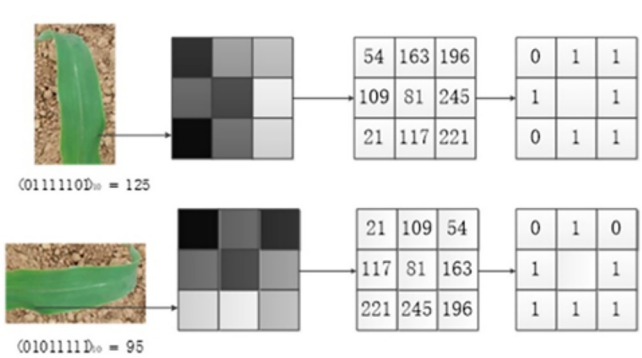

Fig. 6. Process of rotation invariant local binary pattern (RotLBP) (Chen et al., 2020).

In addition to supervised machine learning, weed detection can be accomplished through unsupervised machine learning, specifically useful when dealing with datasets lacking labels. This approach is advantageous in addressing labeling errors (Ota et al., 2022) and unstable identification (Tang et al., 2018) through K-Means clustering. Specifically, the analysis divides plant blobs into two groups, with one representing the crop class and the other the weed class. The distinction is based on the assumption that the quantity of weed is more than those of crop. However, the processing time increases with the number of clusters (Tang et al., 2018), making real-time applications challenging.

To overcome limitations in study data, semi-supervised learning can be used (Khan et al., 2021). Deep Generative Models is a semi-supervised technique that uses the deep learning Generative Adversarial Network (GAN) architecture to model data distribution and optimize model parameters by incorporating both labeled and unlabeled plant data. The semi-supervised learning labeling process is also measured by how fast the data labeling process speed (Lottes and Stachniss, 2017). C. Hu et al. (2021b) use iterative semi-supervised learning to label the data. As a result, the model performance improved. While semantic segmentation can use semi-supervised learning, the accuracy results may not be optimal (Cai et al., 2023). Semi-supervised learning is highly dependent on the quality of unlabeled data. Semi-supervised learning also needs to consider the method's efficiency (K. Hu et al., 2020). Consequently, the efficiency of semi-supervised learning methods should be carefully considered, based on the data's characteristics and the analysis purpose.

5. Weed detection using deep learning

Deep learning has significantly developed, particularly in handling image datasets and achieving high accuracy. In the context of deep learning, CNN has the capability to identify crucial features autonomously during the training phase, reducing the reliance on human intervention for feature extraction (Sarker, 2021). Despite the various advantages, deep learning also has disadvantages, such as requiring large datasets to mitigate the risk of overfitting (Dou et al., 2021). Also, the considerations about hardware usage are crucial for the demanding computational processes (Peng et al., 2022).

Deep learning has received considerable attention in recent years within the broader category of machine learning. Furthermore, deep learning is in line with ANNs, but more closely associated with deeper networks (Nasiri et al., 2022). The unique attribute of the method lies in the capability to automatically learn hierarchical features from raw

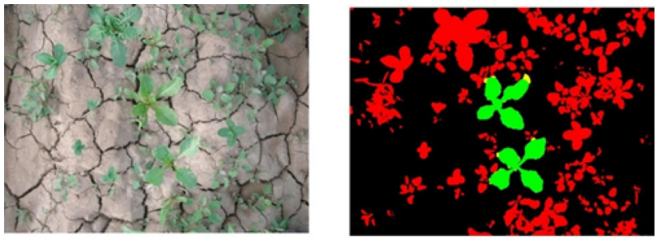

Fig. 7. The result of weed and sugar beet classification (Bakhshipour and Jafari, 2018).

data and perform extraction (Dourado-Filho and Calumby, 2021). According to Fig. 8, the method comprises several stages, namely Data Acquisition, Pre-processing, and Feature Extraction and Classification. The comprehensive evaluation of the strengths and weaknesses of studies using deep learning method is shown in Table 2.

5.1. Pre-processing

In deep learning, the pre-processing stage is an important step in data preparation. For example, when dealing with an insufficient number of datasets with varying sizes and lacking labels, an augmentation technique is needed to improve the dataset (Maram et al., 2022; Tannouche et al., 2022). Data augmentation is performed in the training process to prevent overfitting and compensate for the dynamic nature of target classification. Typically, data augmentation techniques include applying random modifications such as blurring, adding Gaussian noise, adjusting RGB colors, creating shadows, rotating images by 90 degrees, altering brightness and contrast, changing hue and saturation values, and simulating snow or rain effects (Rahman et al., 2023).

The augmentation process can also be carried out by dividing the input image into several patches (Brilhador et al., 2019) and breaking down the image into smaller components, which are processed individually. To ensure uniform-sized sections, a padding procedure fills the borders of the original image with zeros, as shown in Fig. 9. This approach is particularly effective for image datasets with high resolution. Jin et al., 2022b also used grid cells. This approach makes classification more precise for each grid cell in the image. D. Su et al., 2021 used data augmentation strategy, including the Random Image Cropping and Patching (RICAP) method, which significantly increased mean accuracy and intersection over union (IOU). Additionally, resizing is a common practice in deep learning, specifically when implementing the transfer learning method (Ma et al., 2019; Selukar et al., 2022). The operation is important for adjusting the pre-trained data used in transfer learning. For classification requirements, the data labeling process is important, with some opting for manual labeling (D. Hu et al., 2021a), while others use automatic algorithms for large datasets (Bah et al., 2018). The automatic labeling process is optimal for large datasets. The pre-processing stage in deep learning is used as data preparation before subsequent processing.

5.2. Image segmentation

In deep learning, leaf object image segmentation is a technique that groups pixels to distinguish leaves or key features from their surroundings. This process relies on identifying specific attributes such as the color green, texture, and shape of the leaves. While some studies incorporate the segmentation process (Lottes et al., 2018b; Sun et al., 2022), and some do not use the segmentation stages (Dou et al., 2021; Farooq et al., 2018a; Subeesh et al., 2022). Those opting out of segmentation rely on the capabilities of advanced deep learning models, such as CNN, which can autonomously extract crucial features from images, producing accurate results without prior segmentation. However, using the segmentation process enhances deep learning performance and provides precise bounding boxes for plant detection (H. Huang et al., 2018). Fully Convolutional Networks (FCN), a modification of CNN, are optimized for segmentation tasks. By incorporating transfer learning, FCNs demonstrate high accuracy in the semantic segmentation of small plant datasets (Lottes et al., 2018a), as shown in Fig. 10. Specifically, FCN is very useful for image segmentation tasks, specifically when dealing with input images of varying sizes and high precision.

5.3. Feature extraction and classification

Deep learning combines feature extraction and classification into a unified process (Shalaby et al., 2022). In deep learning, the structure of the model is composed of several layers. Each layer is designed to progressively extract and refine more sophisticated features from the input data, enabling the model to understand and represent the data in increasingly complex ways. In addition, the approach allows end-to-end execution of feature extraction and classification processes in deep learning, eliminating the need for manual feature extraction. This implies that deep learning model efficiently and accurately learns crucial features from input data. Various CNN structures have been developed, with some studies using traditional CNN (Farooq et al., 2018a; Gothai et al., 2020; Knoll et al., 2019), while others leverage transfer learning or modifications to existing architectures (Selukar et al., 2022; Tannouche et al., 2022; J. Yang et al., 2022).

Traditional CNN requires building and training the model from scratch to recognize patterns in image data. The model basically requires large datasets (Knoll et al., 2019), and for smaller datasets, data augmentation is needed (Farooq et al., 2018a; Gothai et al., 2020). However, the utilization of traditional CNN often extends the duration of the model training process (Farooq et al., 2018a). On the other hand, Transfer learning leverages a CNN that has already been trained on large datasets to detect patterns within images. The model is then fine-tuned for the specific image data being processed. To achieve more accurate results with less data (Dou et al., 2021; Selukar et al., 2022), transfer learning can be used to accelerate training time (Peteinatos et al., 2020). Typically, transfer learning is very promising in weed detection, contributing to the automation of agricultural processes. In a comparison of three transfer learning methods, such as DenseNet, GoogLeNet, and ResNet, for weed control, Jin et al., 2023 found ResNet to be the most precise, and capable of optimizing herbicide usage on sprayer devices. Another study by Subeesh et al., 2022 compared four transfer learning models, namely Alexnet, GoogLeNet, InceptionV3, and Xception, in identifying weed in a bell pepper field. In terms of precision, accuracy, and recall, InceptionV3 significantly outperformed other models. This result shows that transfer learning ensures accuracy and also enables real-time implementation (Peng et al., 2022; Rahman et al., 2023; H. Zhang et al., 2022a).

The algorithm's speed is dependent on the processing device used. For example, Peng et al., 2022 used NVIDIA GTX TITAN X VGA to achieve

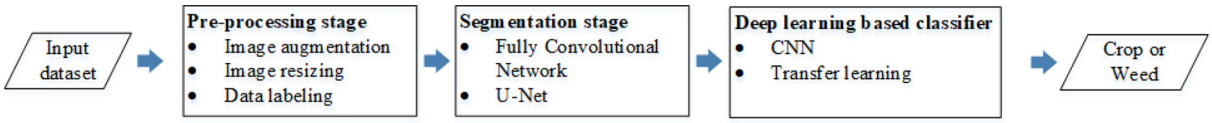

Fig. 8. A workflow of weed detection using deep learning.

Table 2
Strengths and weaknesses of research with the deep learning.

Researcher	Method	Strength	Weakness
Convolutional Neural Network			
Tannouche et al. (2022)	Data augmentation, VGG16, VGG19, InceptionV3, InceptionV4, MobileNetV1, MobileNetV2	The use of transfer learning MobileNetV2 can classify weed and crop.	The segmentation process is performed manually to search for plant ROI.
Maram et al. (2022)	Data augmentation, InceptionV4, EfficientNetB7	The data used varies from several types of soil, namely fertile, clay, and loamy.	The resulting model experiences overfitting.
H. Zhang et al. (2022a)	Data enhancement, EM-YOLOv4-Tiny	The method is accurate and can run in real-time at 10.4 ms per image using the Tesla T4 GPU.	The method is unable to detect weed and crop in early growth.
J. Yang et al. (2022)	VGGNet	Classification in an outdoor environment with lightning clear/bright, cloudy, or partly cloudy skies.	Experiments only on two classes, namely broadleaf weed and grass.
Tiwari et al. (2019)	Resize image, InceptionV2	The method can detect weed from various backgrounds.	Classification of 3 types of weed only.
Dou et al. (2021)	Resize image, flipped and rotated data, MobileNetV2 and Convolutional Block Attention Module (CBAM)	The method is not overfitting for the data classification of 12 classes.	The data is not segmented before the feature extraction process.
Farooq et al. (2018a)	Hyperspectral image with 61 bands, Convolutional Neural	Network Hyperspectral images are better than RGB image-based data in some ways.	There is no segmentation process, hence the image data is in the form of a weed area.
Brilhador et al. (2019)	Patch augmentation, U-Net Architecture	This method is suitable for segmentation processes that separate soil from plants.	Weed and crop classification processes have problems due to visual similarities.
Farooq et al. (2019a)	Multi-Layer Fused Convolution Neural Network (FCNN) and Superpixel based Local Binary Pattern	The method can classify weed and crop with multi-resolution remote sensing images.	The image cropping process is still manual, hence there is no segmentation process.
Badhan et al. (2021)	ResNet-50	Classification results accurately classify weed and onions in different lighting conditions.	The process still uses manual cropping.
Garibaldi-Márquez et al. (2022)	Transfer learning CNN Xception.	Classification at the early stage of plants with natural conditions.	Test time is still not real-time at 144.38 ms (6 fps).
Yu et al. (2022)	Improved DeepLabv3+	Method of solving the edge problem of densely weed and overlap crop.	The time required for the segmentation process is 367 ms.
Peng et al. (2022)	Improved RetinaNet network	The method can run in real-time at 24.3 fps with NVIDIA GTX TITAN X GPU.	The detection process is carried out from the side of the plant object.
Rahman et al. (2023)	YOLOv5n	Using Google Collab, it can reach 17 ms per image	The accuracy obtained is still small.
Knoll et al. (2019)	Convolutional Neural Networks	By using CUDA, the processing time is 31 times faster.	Weed can be grouped into one class.
Subeesh et al. (2022)	InceptionV3	The detection results yield good accuracy for weed and crop in polyhouse classification.	Does not use the segmentation process, and there is no bounding box.
Asad and Bais (2020)	Maximum likelihood classification, ResNet-50	The labeling process uses semantic segmentation in an uncontrolled environment.	If there is an overlap on the leaves, the classification becomes wrong.
D. Hu et al. (2021a)	YOLOv4	Accurate classification results with a total dataset of 22,177 images.	The process cannot run in real time because it requires 377 ms for recognition.
(Selukar et al., 2022)	ResNet-50	The method resulted in an exceptionally low occurrence of both false positive and false negative outcomes.	The feature extraction process does not focus on leaf areas.
Gothai et al. (2020)	Convolutional Neural Network	The validation accuracy value is quite high	The number of datasets is small, and only two classes.
(Peteinatos et al., 2020)	VGG16, ResNet–50, Xception	Very high accuracy for the classification of 12 plant classes	Does not carry out the plant segmentation process
Bah et al. (2018)	Convolutional Neural Network	Use of automatic labeling and no need for feature selection	Background segmentation is not optimal
Razfar et al. (2022)	5-layer CNN architecture	Accuracy and latency are pretty good with Raspberry's implementation	Does not carry out the plant segmentation process
D. Su et al. (2021)	Data augmentation, Bonnet Architecture	The proposed data augmentation method can lead to improved average accuracy as well as a better intersection over union (IOU) metric.	The suggested data augmentation method only works when there are few training images.
Nagothu et al. (2023)	SSD Mobilenet	The use of SSD Mobilenet can detect many objects in the image.	The model tends to overfit because the training accuracy exceeds the testing accuracy.
Jin et al. (2022a)	GoogLeNet, MobileNet-v3, ShuffleNet-v2, and VGGNet	This method can be used for selective herbicide spraying in real-time with automatic smart sprayers.	Using weed control spectrum herbicides can result in misclassification if the area containing weeds has not been detected.
Jin et al. (2022b)	Grid cells, DenseNet, EfficientNetV2, ResNet, RegNet and VGGNet	TA two-classifier neural network is effective in identifying and locating grid cells containing weeds and their species.	Classification is only carried out on four types of weeds.
Fully Convolutional Network			
Sun et al. (2022)	improved region-based fully convolutional network (R-FCN)	The method produces good accuracy in poor light and complex background conditions.	The use of light affects the classification results.
Lottes et al. (2018b)	fully convolutional network and sequential information	This method robustly classifies crops in different growth stages.	The data training process takes 58 h with the NVIDIA GeForce GTX 1080 Ti GPU.
H. Huang et al. (2018)	Fully Convolutional Network and VGG16	FCN accuracy results in weed recognition is higher than other methods.	Processing time per image is quite long, 0.318 s.
Ma et al. (2019)	SegNet method	This method is suitable for pixel-based classification with images of small and	The method requires a longer time to process the image, 0.604 s.

		abnormal plant objects.	
Lottes et al. (2018a)	Fully Convolutional DenseNet	This method properly segments small weed and grass.	The average precision for a crop is still low.
Babiker et al. (2022)	12-layer fully convolutional network	The method can be used on a small amount of data.	The dataset used is manually cropped into sub-images.
Jin et al. (2023)	ResNet	The system has been implemented with a sprayer that can save herbicide due to high precision.	When used on extensive weed, the use of herbicides is also extensive.
Hybrid Network			
Chavan and Nandedkar (2018)	Hybrid AlexNet and VGGNET	Adding the number of classes only slightly affects the training time.	The classification process per image takes 0.97 s.
You et al. (2020)	hybrid dilated convolutional layer and DropBlock	The method can increase the segmentation accuracy	The incorrect annotations significantly impact the performance of the labels.
Al-Badri et al. (2022a)	Hybrid VGG-16, ResNet-50,	Inception-V3 The model can solve various illumination and overlapping conditions on Rumex weed plants.	The model is still implemented on one type of weed.
Nasiri et al. (2022)	Hybrid U-Net and ResNet-50	The model can increase the accuracy of segmentation in small areas.	The overlap between plants and weed causes segmentation faults.
Cheng et al. (2021)	Hybrid MobileNetv3 and YOLOv4	The hybrid method can increase mAP and detection speed.	The resulting map is still below 90% and uses CUDA configuration.

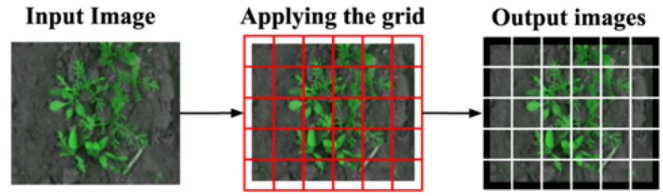

Fig. 9. Illustration of patch augmentation (Brilhador et al., 2019).

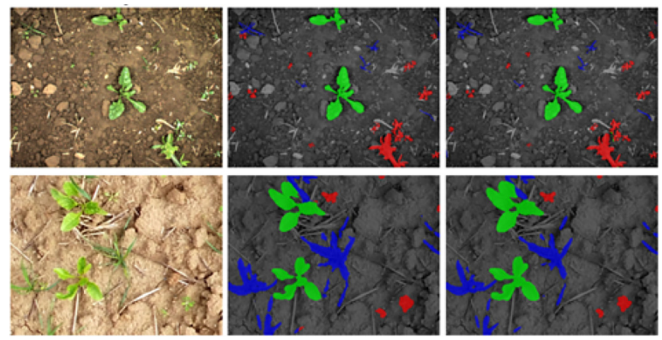

Fig. 10. Results of semantic segmentation (Lottes et al., 2018a).

a speed of 24.3 fps for weed classification in rice fields. Similarly, H. Zhang et al. (2022a) used a Tesla T4 GPU with VGA, resulting in a real-time performance of 10.4 ms or 96 fps for weed classification in peanut fields using EM-YOLOv4-Tiny. The outcomes of weed detection in peanut fields can be seen in Fig. 11. The hybrid deep learning approach further enhances speed, supported by a computer with VGA NVIDIA GeForce GTX 1080Ti (Cheng et al., 2021), producing an impressive 69.76 fps in experimental results. The implementation of quantization in deep learning is another technique to increase speed (Razfar et al., 2022), simultaneously reducing file size and minimizing latency. Based on the capability for real-time operation, computer vision algorithms can be easily integrated into compact devices, thereby proving valuable across various agricultural aspects, including large farms. Consequently, the integration aids farmers in increasing productivity and effectively managing weed control on the farm.

6. Weed detection using hybrid method

Integrating CNN with machine learning significantly improves classification accuracy. CNN extracts complex features from images through convolution and activation functions across multiple layers, while machine learning utilizes these features for precise classification. After the features are extracted, machine learning can be used for classification. Consequently, the extracted features can be used as input for a machine-learning model. The strengths and weaknesses of the hybrid method are shown in Table 3.

Fig. 11. Weed detection results in peanut fields using EM-YOLOv4-Tiny (H. Zhang et al., 2022a).

Table 3
Strengths and weaknesses of research with the hybrid method.

Researcher	Method	Strength	Weakness
Julie et al. (2021)	Region-Based CNN with Support Vector Machine	The use of bounding box results from region-based CNN produces good accuracy.	Long training process with few datasets.
Tao and Wei (2022)	VGG Net - SVM	The accuracy results are quite good at three times, morning, noon, and afternoon.	The training process looks overfitting.
Kempegowda et al. (2022)	Handcrafted and CNN, Bayesian-SVM	The resulting accuracy is quite good.	The dataset consists of only two classes.
Espejo-Garcia et al. (2020)	DenseNet - SVM	The transfer learning method does not occur the overfitting	Not all transfer learning results in optimal accuracy.
Jiang et al. (2020)	VGG16 - SVM	The method produces good accuracy for classifying weeds and plants in early growth.	The dataset used is only a single plant for each image.

Machine learning model can be trained to use classification algorithms, such as SVM. The integration of a Hybrid CNN and SVM approach produces more precise classification results. In a study by Julie et al., 2021, a region-based CNN (RCNN) coupled with SVM is used to identify weed in a shared field. Consequently, the classification results show that as the epoch value increases, the resulting model's accuracy improves, reaching a peak of 95.66%. Another study by Kempegowda et al., 2022, using a combination of handcrafted and CNN features, achieves 100% accuracy in weed detection on carrot farms. When combined with SVM, the transformation of CNN into DenseNet transfer learning model effectively prevents overfitting (Espejo-Garcia et al., 2020). Empirical observations suggest that plant segmentation reduces overfitting, while the usage of a fine-tuning technique enhances performance. The application of transfer learning with SVM also improves accuracy under specific lighting conditions (Tao and Wei, 2022) and with early-growth plants (Jiang et al., 2020). However, it should be acknowledged that the combination of CNN and machine learning requires more computational time and resources.

7. Performance method using dataset benchmark

In this research, the benchmark dataset will discuss the results of previous research using the rice and corn dataset. The results of a review of various types of weeds were also added to see the accuracy of the method in detecting weeds. Benchmark this dataset to determine which method is the most powerful in detecting weeds.

7.1. Weed in rice and corn fields

The selection of datasets from rice and corn fields as benchmark data for weed and crop classification is based on several factors, as shown in Table 6. For example, rice and corn are important crop commodities in Indonesia and various countries worldwide (Kamaludin et al., 2021). Therefore, dataset classification can assist in the development of precision agricultural technology as an effort to control weed around rice and corn plants. Rice and corn have different shapes and characteristics depending on the type and age (D. Li et al., 2023; W. Su et al., 2018). The datasets from paddy fields and corn fields served as perfect testing grounds for the model's proficiency in classifying objects with diverse characteristics. Consequently, selecting datasets from rice and corn fields as a case study for weed control via camera sensors offers multiple advantages in supporting the development of agricultural technology, facilitating the transition from traditional to modern methods. The benchmark datasets for corn and paddy are shown in Tables 4 and 5, respectively.

Weed has the potential to thrive around various agricultural crop. This benchmark dataset discusses the classification outcomes for corn and rice in the presence of weed. By using machine learning in combination with rotation-invariant LBP and GGCM, SVM achieves an impressive accuracy of 97.5 (Chen et al., 2020). The training process is relatively fast, with 2.1903 s to train 1400 images. Additionally, the computer specifications include a core i7-9750H @ 2.60GHz X 12, GeForce RTX 2060 with 16 GB memory. Subsequently, using a hyperspectral camera produces a denser image, incorporating more features. However, when classified with Random Forest, the effect of many of the features produces a suboptimal F-measure (Gao et al., 2018). This implies that machine learning may be inaccurate in situations where there are overlapping plant leaves. Moreno and Cruvinel, 2022 used SVM with an accuracy of 72.2%, indicating the presence of overlapping obstacles in the leaves.

Apart from machine learning, some studies have explored deep learning for weed classification with corn plants. Deep learning leverages pre-trained models where the dataset has been previously trained. An example of the model is YOLO, which is used for the classification of weed and maize (Cheng et al., 2021; B. Wang et al., 2023; Zhu et al., 2022). The classification results show that the accuracy has not yet reached 90%. However, in the research conducted by Cheng et al., 2021, the method achieves real-time results of 69.76 fps with a parallel GPU GeForce 1080Ti. Transfer learning models such as ResNet (Jiang et al., 2020) and VGG16 (Ni et al., 2022; Sunil et al., 2022) are pre-

Table 4
Corn dataset.

Researcher	Method	Code	Example plant	Acc (%)
Chen et al. (2020)	rotation-invariant LBP features and GGCM, SVM	A		97.5
Gao et al. (2018)	Normalized difference vegetation index (NDVI), ratio vegetation index (RVI), Band reflectance, PCA, Random Forest	B		80.8
Moreno and Cruvinel (2022)	HSV with green Hue, shape, area, perimeter, width, height, proportion, rectangularity, circularity, Haralick moment, SVM	C		72.2
Jiang et al. (2020))	Hybrid GCN-ResNet-101	D		99.25
Diao et al. (2022)	lightweight-3D-CNN	A		98.58
Sunil et al. (2022)	VGG16	E		97.32
Ni et al. (2022)	VGG16	F		96.2
B. Wang et al. (2023)	improved yolov5s	A		83
Zhu et al. (2022)	YOLOX convolutional neural network	A		88.94
Cheng et al. (2021)	MobileNetv3 - YOLOv4	G	–	89.98

Table 5
Paddy dataset.

Researcher	Method	Code	Example plant	Acc (%)
Kamath et al. (2020)	Color features, size-independent features, Hu's moments, Laws' texture masks, Multiple classifier systems (support vector machines and random forest)	H		91.36
Dadashzadeh et al. (2020)	Artificial neural network - bee algorithm	I		92.02
Kawamura et al. (2021)	HSV and texture combination, simple linear iterative clustering (SLIC) algorithm, and random forest (RF)	J	–	91.5
Peng et al. (2022)	Improved RetinaNet network	K	–	94.1
Zamani and Baleghi (2023)	Morphological, spectral, textural, thermal features, genetic algorithm, extreme learning machine	J		98.08
Kamath et al. (2022)	Pyramid Scene Parsing Network (PSPNet) - ResNet-50	L		90
Yan et al. (2020)	Deep Convolutional Neural Networks	M		94.5
D. Hu et al. (2021a)	YOLOv4	N		97

trained CNN models applied to address this case study. Consequently, the transfer learning model produces excellent accuracy of more than 97%. In line with the specifications of GeForce GTX 1080 Ti alongside an Intel Core i9-7920× CPU operating at 2.9 GHz, Jiang et al., 2020 adopted a method that run in real-time and processes 1800 images in only 1.24 s. This signified that both machine learning and deep learning can produce optimal accuracy. However, it is essential to use a computer with good specifications for real-time applications.

Machine learning classification algorithms often require intensive computations, such as matrix multiplication or convolution. To ensure efficient parallel processing of the operations, GPUs need to be used with CUDA technology. When rice plant is classified with weed, machine learning achieves high accuracy above 90% (Dadashzadeh et al., 2020; Kamath et al., 2020; Kawamura et al., 2021). However, a combination of ANN with Bee algorithms is impractical for real-time processing (Dadashzadeh et al., 2020). This limitation arises due to the use of computers equipped with an Intel Core i5-2540m processor running at 2.6 GHz, coupled with 4 GB of RAM, and lacking a dedicated GPU. A combination of several features from visual and thermal images is also very accurate for detecting weeds around rice plants (Zamani and Baleghi, 2023). Even the dataset used in this research uses real-world conditions. Additionally, the current review is the application of deep learning for rice classification with weed. YOLOv4 (D. Hu et al., 2021a) and RetinaNet modification (Peng et al., 2022) demonstrate commendable accuracy at 97% and 94.1%, respectively. The study by Peng et al., 2022 describes real-time processing at 24.3 fps with an NVIDIA GTX TITAN X GPU. In summary, deep learning generally achieves higher accuracy than machine learning, but in the aspect of speed, machine learning performs better in real-time processing.

7.2. Vast diversity of weed species

Various weed species have different leaf shapes and textures. This diversity of species can make it difficult to detect the presence of weeds around cultivated plants. Table 7 shows previous studies that used datasets of several types of weeds.

From previous research on case studies of weed species diversity, all methods use deep learning. Two methodical approaches to identifying weeds are identifying each weed species and ignoring the weed species. Research by Ni et al., 2022 and Yan et al., 2020 classified six species of weeds and their crops with good accuracy results. These two studies did not carry out a segmentation stage on their datasets, so it can be said that the deep learning method without segmentation is powerful enough to classify the diversity of weed species. Then, research by Sunil et al., 2022 classified ten types of plants consisting of weeds and crops, which also produced very good accuracy. In this study, background removal or segmentation of the leaf area was not carried out. Other researchers identified 12 species consisting of weeds and crops (Dou et al., 2021; D. Hu et al., 2021a; Peteinatos et al., 2020). The research results show that it is very accurate in identifying weed types, and the resulting model does not occur overfitting. Other research in identifying various weed species focuses on vegetable plants so that other green leaf objects are classified as weeds (Jin et al., 2022c; Jin et al., 2021). Both studies also used deep learning to detect vegetables,

Table 6
Code of weed and crop dataset.

Plant	Code
Corn seedlings and weeds	A
Convolvulus arvensis, Rumex, *Cirsium arvense*, and *Zea mays*	B
Zea mays L., Euphorbia heterophya L., and Cenchurus echinatus L.	C
Corn seedlings, *Cirsium setosum*, *Chenopodium album*, Bluegrass, and Sedge	D
Black bean, Canola, Corn, Flax, Soybean, Sugar beet, Horseweed, Kochia, Ragweed, Waterhemp	E
Black-grass, Charlock, Cleavers, Common chickweed, Loose silky-bent, and Maize seedlings	F
Maize seedlings, dandelion, acalypha, bowl flower, and *convolvulus arvensis*	G
Paddy, Broadleaf weeds, Grass, and Sedges	H
Rice, Narrow Weeds, Broadleaf weeds	I
Iranian paddy, cockspur grass	J
Rice, Ludwigia prostrata Roxb, *Leptochloa chinensis*, *Monochoria vaginalis*, Alternanthera philoxeroide, Sagittaria pygmaea Miq, *Echinochloa crusgalli*, *Portulaca oleracea* L, *Cyperus rotundus* L	K
Paddy, Broadleaf weeds, Grass, Sedge	L
Alternanthera philoxeroides, *Eclipta prostrata*, Ludwigia adscendens, *Sagittaria trifolia*, *Echinochloa crusgalli*, and *Leptochloa chinensis*	M
Cyperus alba, Polygonum syringae, Amaranthus multiflora, *Echinochloa crusgalli*, *Alternanthera philoxeroides*, *Echinochloa crusgalli*, Chanchus Argus, Oxtendon grass, Euphorbia, Cyperus fragrans, *Echinochloa crusgalli* and Weedy rice	N
Black-grass, Charlock, Cleavers, Common chickweed, Common wheat, Fat hen, Loose silky-bent, Maize, Scentless mayweed, Shepherd's pure, Small-flowered cranesbill, Sugar beet	O
Maize, Potato, Sunflower, *Alopecurus myosuroides* Huds, *Amaranthus retroflexus* L., Avena fatua L., Chenopodium album L., Lamium purpureum L., Matricaria chamomila L., Setaria spp., *Solanum nigrum* L., Stellaria media Vill	P
Bok choy and various weed species	Q

Table 7
Dataset with diverse weed species.

Researcher	Method	Code	Acc. (%)/F_1
Sunil et al. (2022)	VGG16	E	Acc. 97.32
Ni et al. (2022)	VGG16	F	Acc. 96.2
Yan et al. (2020)	Deep Convolutional Neural Networks	M	Acc. 94.5
D. Hu et al. (2021a)	YOLOv4	N	Acc. 97
Dou et al. (2021)	MobileNetV2 and Convolutional Block Attention Module (CBAM)	O	Acc. 96.88
Peteinatos et al. (2020)	Xception	P	Acc. 98
Jin et al. (2021)	CenterNet, color index, Genetic Algorithm, Bayesian Classification	Q	F_1 0.953
Jin et al. (2022c)	YOLO-v3, excess green (ExG), Otsu	Q	F_1 0.971

so the main focus was to detect vegetables first. This strategy can reduce the complexity of weed detection. The experimental results also show very good F_1 values. Overall, various weed species can be identified using deep learning for accurate results. GPU devices have also supported deep learning to speed up the training and testing.

8. Future work and discussion

Weed and crop detection had several challenges that required attention, as shown in Fig. 12. The first challenge included non-uniform lighting effects, which could change the contrast and create confusing shadows, thereby impacting detection. Second, leaf occlusion between weed and crop could lead to difficulty in differentiation. Additionally, weed detection in the early stages of crop growth was challenging due to the visual similarity. Lastly, achieving real-time detection necessitated a method with rapid processing responses. These challenges were explored and then the potential solutions were discussed.

8.1. How to detect weed and crop with lighting effect

Lighting influenced by weather conditions could significantly affect weed detection (J. Yang et al., 2022). Additionally, insufficient or excessive lighting could lead to difficulty in distinguishing weed from crop, leading to inaccurate detection. In artificial intelligence, image processing relied on clear and high-resolution data for precise object differentiation (Alam et al., 2020). Therefore, poor lighting might result in blurry or dark images, impeding algorithms in distinguishing between weed and plant. Similarly, overexposure could create an overexposed image, complicating weed detection and causing shadows between leaves (Farooq et al., 2019b).

The combination of various features could minimize the impact of lighting. Alam et al., 2020 used the Random Forest method with shape, texture, and color features to effectively classify weed and crop. It should be acknowledged that clustering techniques offered another solution (Tang et al., 2018). Therefore, increasing clustering numbers and exhaustive feature extractions could improve the network's identification rate. Another approach for addressing the lighting challenge is transfer learning. J. Yang et al., 2022 applied VGGNet to classify weed in alfalfa plants across outdoor environments with varying lighting conditions. Similarly, Asad and Bais, 2020 achieved successful weed classification in canola fields using transfer learning ResNet-50. This signified that the combination of different features in machine learning allowed the model to learn from various leaf image variations, thereby rectifying accuracy weaknesses caused by lighting issues. To improve accuracy, transfer learning leveraged existing knowledge by varying the dataset under different lighting conditions during training.

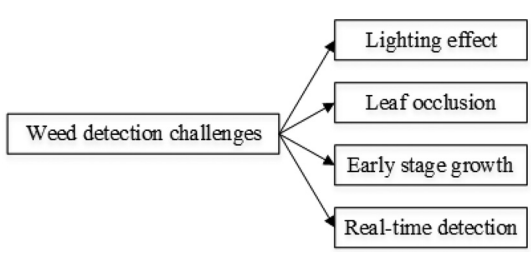

Fig. 12. Weed detection challenges.

8.2. Detection of weed and crop in occlusion conditions

Occlusion known as overlapping objects in an image significantly impacted weed and crop detection, particularly during the identification process (Zhao et al., 2022). Furthermore, occlusion occurred when an object, such as weed or crop obscured another object or plant. This scenario caused difficulty for object detection algorithm to distinguish between weed and crop because the object that should have been visible in the image was covered. Object detection algorithms identified specific objects by finding patterns and special features in images. When objects in the image were obstructed or truncated, object detection algorithms struggled to find the necessary patterns or special features for accurate object recognition, particularly when using visual features (Moreno and Cruvinel, 2022).

Occlusion on leaves could be addressed using edge detection and thresholding. Methods such as Otsu's thresholding and Canny's edge detection could resolve overlapping issues in plants (Ota et al., 2022). The use of the Bag of Visual Words augmented and Histogram of Oriented Gradients feature extraction method could also address leaf overlapping (Abouzahir et al., 2021). The results showed that shape features and feature descriptors effectively resolved the problem of overlapping leaves. Additionally, the utilization of semantic segmentation resolved overlapping issues. The Swin-DeepLab architecture employs a Swin transformer block as its primary mechanism for backbone feature extraction, moving away from the traditional convolution operation. This process allowed the model to extract more distinct feature information, resulting in improved segmentation of the overlapped section (Yu et al., 2022). The use of hybrid transfer learning was also effective in addressing leaf occlusion (Al-Badri et al., 2022a). However, it should be acknowledged that not all deep learning approaches could solve the occlusion problem (Asad and Bais, 2020; Nasiri et al., 2022). To separate occluded leaves, a good segmentation technique would be required to accurately perform feature extraction.

8.3. Classification of weed and crop in early stage

Weed detection with small seed caused a significant challenge (Ota et al., 2022). This issue arose due to the small size of the seed, making it easily concealed by weed, and often the sizes were nearly identical. For example, when capturing an aerial photo, it was difficult to differentiate between crop and weed, as they have the same green color, and the leaves were similar in size (Brilhador et al., 2019). Consequently, special techniques were essential for effectively classifying weed from crop.

Machine learning and deep learning offered methods for classifying early-growth weed and crop. For example, SVM machine learning could appropriately classify sugar beet and weed (Bakhshipour and Jafari, 2018). Additionally, shape information was used as a feature, which

enabled SVM to use vectors of physical characteristics. SVM was also effective in classifying seedling corn and weed (Chen et al., 2020), meanwhile, optimal accuracy was achieved by combining two feature extraction methods, namely rotation-invariant LBP features and GGCM. Similarly, FCN could classify small weed and crop through a robust segmentation method. FCN is useful in classifying plants at various growth stages, specifically when coupled with sequential information (Lottes et al., 2018b). Studies had addressed the classification challenge by using sequential information to determine the pattern of crop layout. Furthermore, FCN could be combined with transfer learning for plant segmentation processes. Lottes et al. (2018a) used the Fully Convolutional DenseNet method to segment four classes, namely crop, dicot weed, grass, and soil. This method effectively segmented small weed and grass, while FCN adopted CNN for each image pixel, generating a small object detection map. Aside from this outcome, machine learning could classify small plants with the appropriate feature extraction method.

8.4. Real-time weed and crop detection for smart farming

Real-time weed and plant detection using an artificial intelligence-based sprayer represented an innovative and effective solution to increase agricultural productivity and achieve more efficient weed control. In this system, a camera or sensor was installed on the sprayer, enabling the control of weed and herbicide spraying (Alam et al., 2020). Speed and accuracy were crucial to ensure that the sprayer effectively targeted weed while avoiding damage to healthy plants. Consequently, image detection and processing were carried out swiftly and precisely. Smart farming-based sprayers using Artificial Intelligence for weed and plant detection could significantly improve weed control efficiency, speed, and the total cost-effectiveness of herbicides (S. Zhang et al., 2021). In the long term, this technology could aid farmers increase farm productivity and reduce manual weed control costs.

Some methods had high accuracy, but their complexity could prevent real-time processing. Dadashzadeh et al., 2020 used a combination of Artificial Neural Network-bee algorithm methods, resulting in prolonged computation times without GPU utilization in the classification process. Another instance of accurate results came from Chavan and Nandedkar, 2018, who used a combination of AlexNet and VGGNET transfer learning. However, despite achieving good accuracy, the processing time for classification was below real-time, even with high-performance computers. An example of the approach considered to enable real-time processing was to parallelize computer programs using GPU (Umamaheswari et al., 2018). Ahmad et al., 2018 resized the video frame size to 240 × 320, allowing the program to run at 28.4 fps. Additionally, Knoll et al., 2019 used parallelism with VGA CUDA for real-time classification using the CNN method. Researchers Jin et al. (2022a); Jin et al. (2022b)also used GPUs to detect weeds in turfgrass, where the inference time method can run in real-time. The use of herbicide weed control spectrum (Jin et al., 2022a) and grid cells (Jin et al., 2022b) is also very good, as the resulting accuracy is also very good. So, this method can be used as a smart sprayer based on machine vision. This deep learning method could also be modified for real-time processing. Peng et al., 2022 adapted RetinaNet to achieve real-time operation, with the program running at 24.3 fps. The problem of limited device usage was addressed by quantizing deep learning to achieve real-time (Razfar et al., 2022). Furthermore, quantization methods reduced the sizes of deep learning models, enabling the system to run in real-time. These strategies, including parallel computing, quantization, and method modifications, were crucial for achieving real-time processing.

9. Conclusion

In conclusion, this study provided a comprehensive review of the application of traditional machine learning, deep learning, and hybrid approaches in the detection and classification of weed surrounding agricultural crop. The analysis included 81 papers, identifying their respective strengths and weaknesses. Furthermore, the study explored pre-processing, data segmentation, feature extraction, and classification methods, and offered detailed insights into each aspect. Many publications described the benefits of both machine learning and deep learning. Specifically, machine learning required a method for extracting characteristics, especially those related to leaf objects, which could be achieved through visual features, statistics, or descriptors. Selecting the right features as input for machine learning methods was essential for accurate classification, particularly under varying lighting conditions and during the early stages of weed-crop growth. In cases of occlusion, a segmentation stage had to be performed before the feature extraction process in order to separate leaf objects experiencing occlusion.

In deep learning approach, feature extraction and categorization were automatically conducted on crop and weed leaves. Although some studies lacked a leaf area segmentation procedure, the use of CNN technology in deep learning consistently produced accurate results. Despite the accuracy in classification under different conditions, deep learning has limitations in the aspect of high hardware requirements, thereby requiring resources such as a GPU. However, the development of GPU devices continues to increase, making them easy for researchers or developers to obtain. Compared with machine learning, which usually does not use a GPU device, programmers can select an accurate feature extraction approach for real-time execution based on plant features. To achieve real-time data processing, various methodologies were used for the deployment of small devices coupled with sprayers for herbicide application. Strategies such as parallel processing, quantization of the produced model, and changes in feature extraction in deep learning, alongside traditional machine learning producing small-sized models, could be implemented to attain real-time speed. An important avenue for future exploration comprised integrating the strategies to achieve real-time and precise weed control. Additionally, exploring the adoption of the strategies for farmers through the development of cost-effective precision pesticide sprayers presented another potential area for future investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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